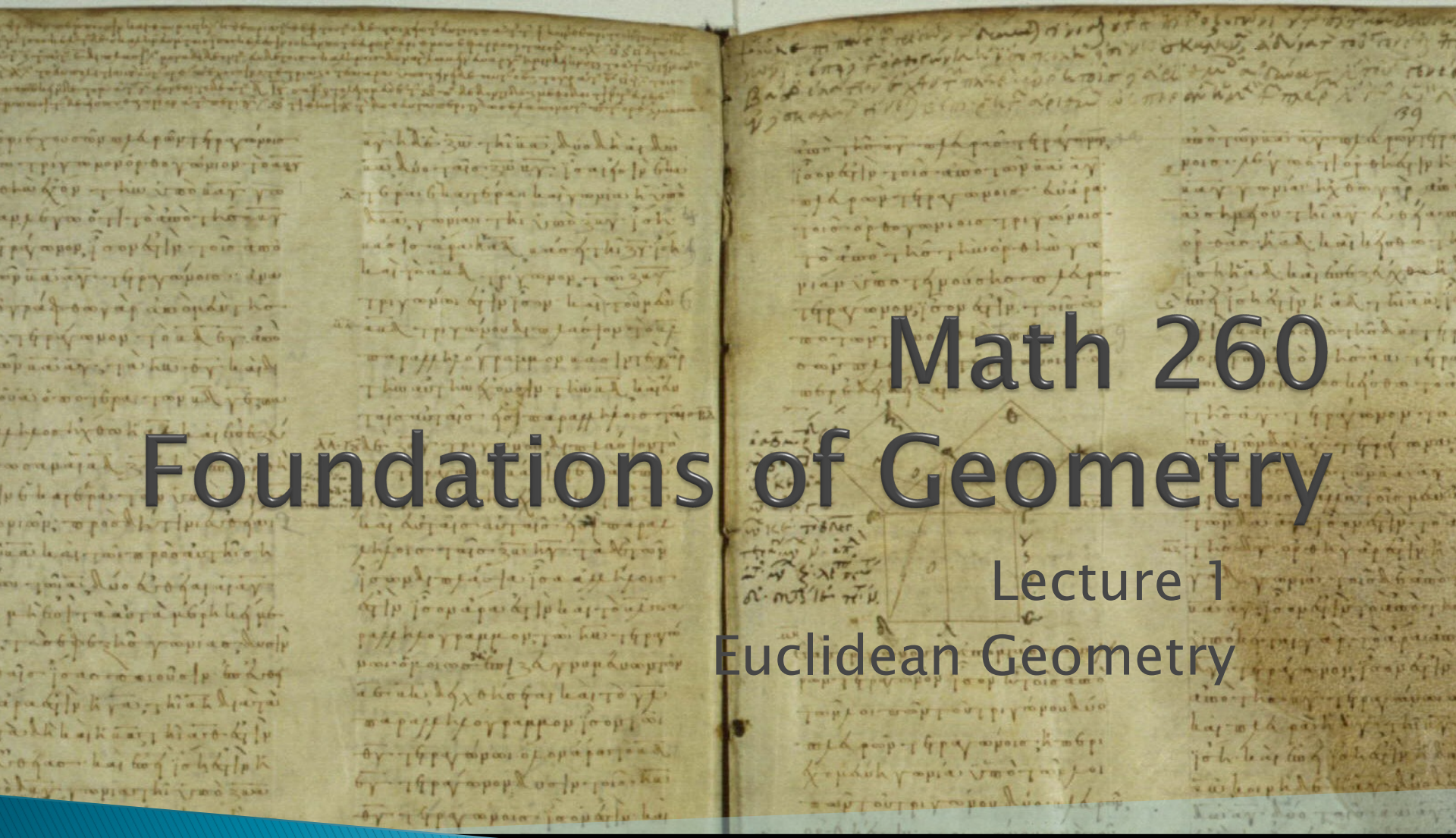


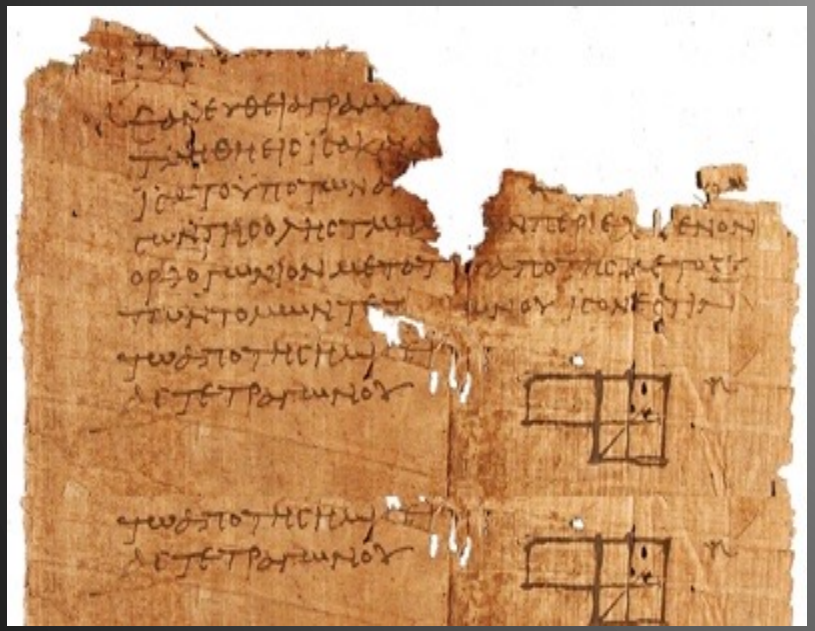
Math 260
Foundations of Geometry

Lecture 1
Euclidean Geometry



- From Alexandria Egypt
- Greek Mathematician
- “Father of Geometry”
- Wrote the famous book Elements
- Also wrote works on conic sections, and number theory

Elements



- Books I-IV and VI discuss plane geometry
- Books V and VII-X deal with number theory
- Books XI-XIII concern solid geometry
- The first mathematics book to be translated by Arabs in Beit Al Hikma, Bagdad

Section 1.2

Define the following notions:

Point?

line ?

Segment ?

Congruence ?

Definitions

- A Point is that has no part.
- A line is breadthless length.
- Straight line?
- Segment?
- plane angle?
- right angle, acute angle, obtuse angle
- Circle, center, radius, diameter

Definitions

- Triangles
- Equilateral triangle,
- Right angled triangle
- Rectangle, square
- Parallel lines: two lines that don't meet
- perpendicular lines

……

Euclid

Around 300BC, the Greek mathematician **Euclid** wrote a set of 13 books called Elements. He listed 5 *common notions* and 5 **postulates** from which everything else can be developed. Indeed all of Euclid's 465 propositions are proved from these 10 facts.

The 5 common notions have logical base – for example

$x=y \Rightarrow x+z = y+z$ for all numbers z – while the 5 postulates are geometric:

Postulate (Axiom)?

- **Postulate:** A statement that we accept as true without justification.

Example: Given a line D and a point P: There is a unique line through P and parallel to D.

Can you prove this "obvious" fact?

Euclid's 5 Postulates

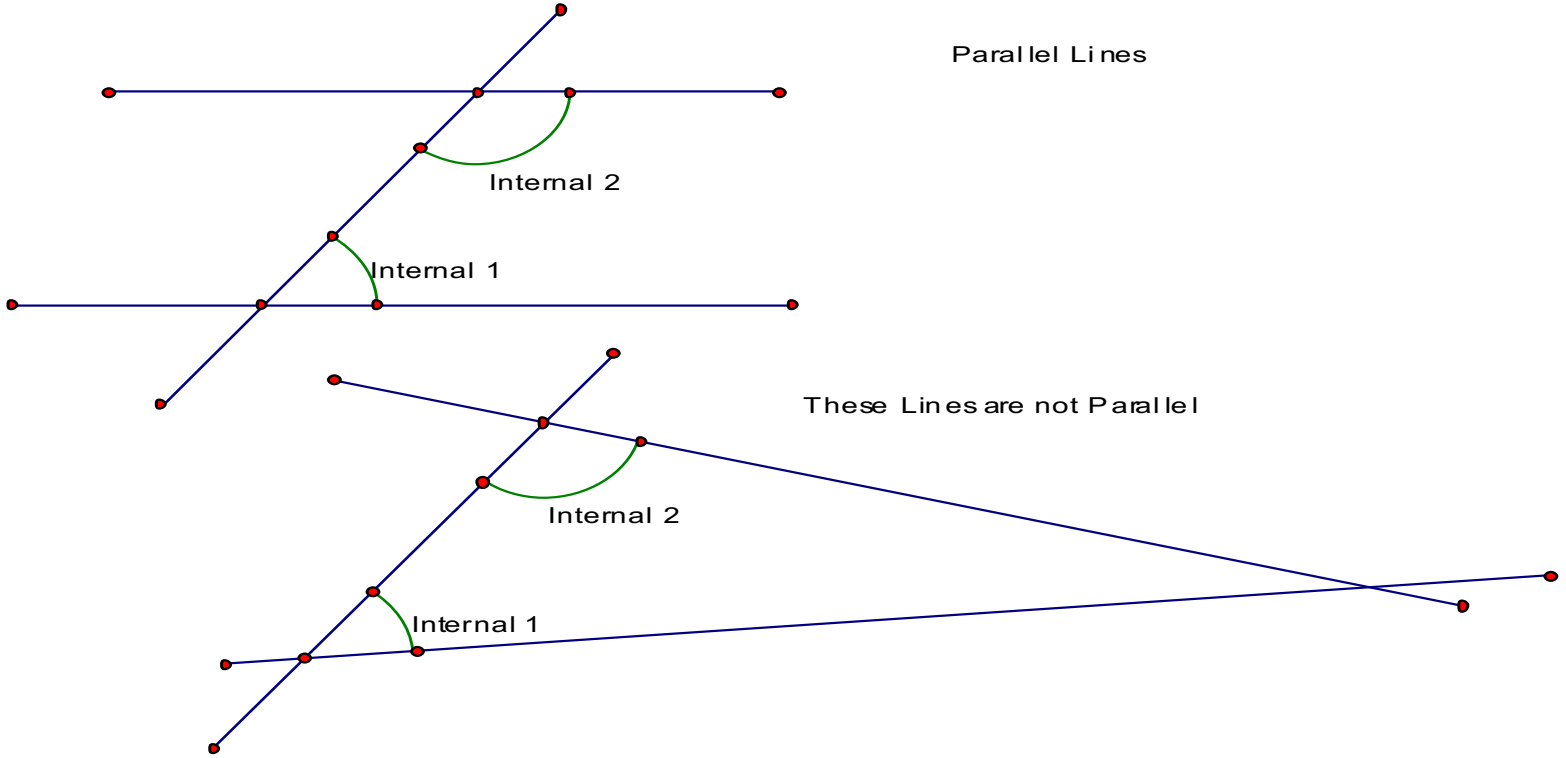
1- A line can be drawn from any point to another.

2- A finite line can be extended indefinitely.

3- A circle can be drawn centered at any point and with any chosen radius.

4- All right angles are equal to one another.

5. If a straight line *D* falls on two lines *D1* and *D2* in such a way that the interior angles on one side of *D* are less than two right angles, then the lines will meet on that side.



The fifth postulate

- Thankfully, the 5th postulate is equivalent to another that is simpler to state.

- <u>The Parallel Postulate (John Playfair 1795)</u>
  - Given a line *D* and a point P not on that line, there is exactly one line through P that is parallel to *D.*

Common Notions

1. Things that equal the same thing also equal one another.
2. If equals are added to equals, then the wholes are equal.
3. If equals are susbtracted from equals, then the remainders are equal.
4. Things that coincide with one another equal one another.
5. The whole is greater than the part.

Conclusion

- All the results in Euclidean Geometry you know can be proved starting from the 5 Postulates, the common notions and some basic definitions : Point, Line, Circle ….

First Proof in Elements

- Prove that you can construct an equilateral triangle from a finite straight line.
- Given: Let AB be the given finite straight line.
- **Hint: Construct two circles of diameter AB.**